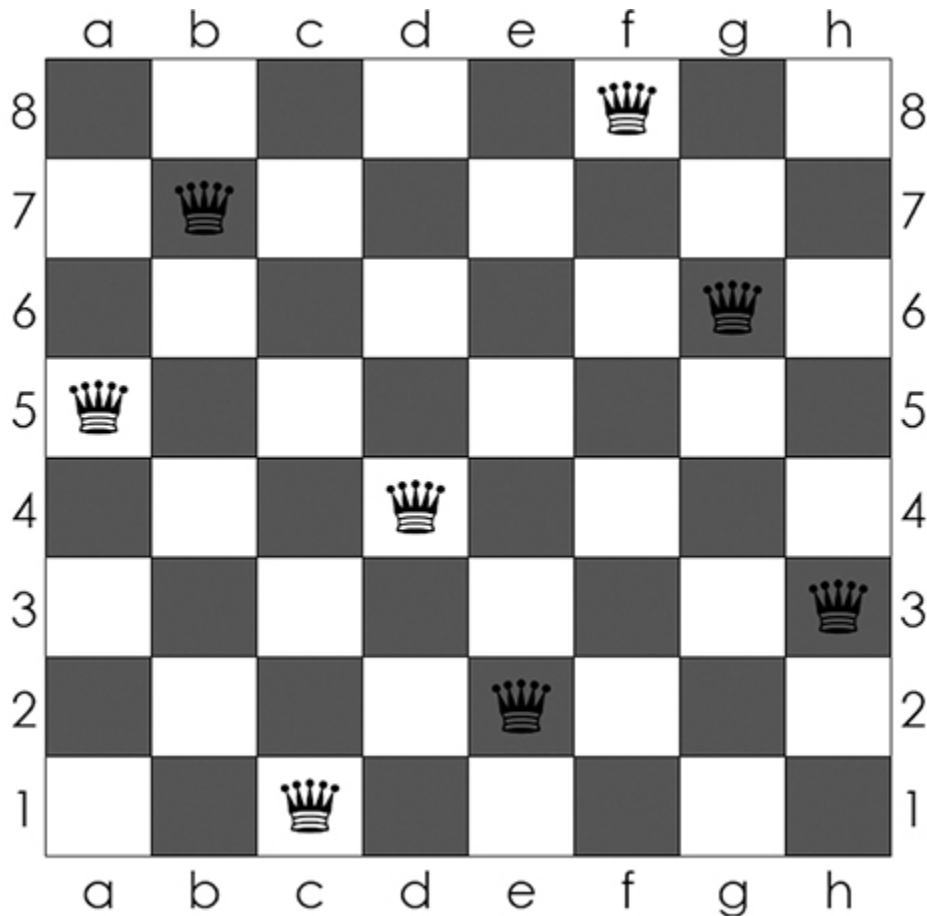




[Figure 11-1](#): A solution to the eight queens puzzle (source: Wikimedia Commons)

None of the queens on this board attacks any of the other queens. The easiest possible way to solve the eight queens puzzle is to simply memorize a solution, like the one in [Figure 11-1](#), and repeat it whenever you're asked to solve the puzzle. However, a couple of extra twists to the puzzle make memorization infeasible. One twist is to increase the number of queens and the size of the board. The *n queens problem* asks us to place n queens on an $n \times n$ chessboard such that no queen attacks any of the others; n could be any natural number, no matter how high. Another twist is the *n queens completion problem*: your opponent starts by placing some of the queens, maybe in places that will make it difficult for you to place the rest, and you have to place the rest of the n queens so that none attack any others. Can you design an algorithm that will run very quickly and solve this problem? If so, you could earn a million dollars (see "Solving the Deepest Mysteries" on page 212).